Since for the irreversibility of the exchange of energy between an oscillator and the radiation activating it, the second differential quotient of its entropy with respect to its energy is of characteristic significance, I calculated the value of this function on the assumption that Wien's Law of the Spectral Energy Distribution is valid—a law which was then in the focus of general interest; I got the remarkable result that on this assumption the reciprocal of that value, which I shall call here *R*, is proportional to the energy. This relationship is so surprisingly simple that for a while I considered it to possess universal validity, and I endeavored to prove it theoretically. However, this view soon proved to be untenable in the face of later measurements. For although in the case of small energies and correspondingly short waves Wien's Law continued to be confirmed in a satisfactory manner, in the case of large values of the energy and correspondingly long waves, appreciable divergences were found, first by Lummer and Pringsheim; and finally the measurements of H. Rubens and F. Kurlbaum on infrared rays of fluorspar and rock-salt revealed a behavior which, though totally different, is again a simple one, in so far as the function *R* is proportional not to the energy but to the square of the energy for large values of the energy and wave-lengths.

Thus, direct experiments established two simple limits for the function *R:* For small energies, *R* is proportional to the energy; for larger energy values *R* is proportional to the square of the energy. Obviously, just as every principle of spectral energy distribution yields a certain value for *R*, so also every formula for *R* leads to a definite law of the distribution of energy. The problem was to find such a formula for *R* which would result in the law of the distribution of energy established by measurements. Therefore, the most obvious step for the general case was to make the value of *R* equal to the sum of a term proportional to the first power of the energy and another term proportional to the second power of the energy, so that the first term becomes decisive for small values of the energy and the second term for large values. In this way a new radiation formula was obtained, and I submitted it for examination to the Berlin Physical Society, at the meeting on October 19, 1900.

The very next morning, I received a visit from my colleague Rubens. He came to tell me that after the conclusion of the meeting he had that very night checked my formula against the results of his measurements, and found a satisfactory concordance at every point. Also Lummer and Pringsheim, who first thought to have discovered divergences, soon withdrew their objections; for, as Pringsheim related it to me, the observed divergences turned out to have been due to an error in calculation. Later measurements, too, confirmed my radiation formula again and again, —the finer the methods of measurement used, the more accurate the formula was found to be.

But even if the absolutely precise validity of the radiation formula is taken